

PSLE MATHEMATICS PRACTICE PAPER 2

Geometry, Angles & Lines

Duration: 1 hour

Total Marks: 50

Name: _____

Date: _____

SECTION A: Short Answer Questions (20 marks)

1. Find the sum of angles in a triangle. (2 marks)

2. If one angle in a triangle is 90° , and another is 35° , find the third angle. (2 marks)

3. A straight line has an angle of 180° . If one angle is 65° , what is the other? (2 marks)

4. Name 3 types of triangles based on angles. (3 marks)

5. What is the sum of all interior angles in a quadrilateral? (2 marks)

6. If a rectangle has length 8 cm and width 5 cm, what is its perimeter? (2 marks)

7. Define a perpendicular line. (2 marks)

8. Two parallel lines are cut by a transversal. Name one pair of equal angles. (3 marks)

SECTION B: Long Answer Questions (30 marks)

9. The angles of a quadrilateral are in the ratio 2:3:3:4. Find each angle. (4 marks)

10. A triangle has angles $2x$, $3x$, and $4x$. Find the value of x and each angle. (4 marks)

11. In a parallelogram, one angle is 70° . Find all four angles. (3 marks)

12. A square has a side length of 6 cm. Calculate: (a) Perimeter, (b) Area (3 marks)

13. Two angles on a straight line are $(3x + 10)^\circ$ and $(2x - 10)^\circ$. Find x and both angles. (4 marks)

14. An isosceles triangle has two equal angles of 45° each. Find the third angle. (2 marks)

15. Draw a diagram of vertically opposite angles and explain why they are equal. (3 marks)

SOLUTIONS

Q1: Sum of angles in triangle

$$180^\circ$$

Q2: Find third angle

$$180^\circ - 90^\circ - 35^\circ = 55^\circ$$

Q3: Straight line angle

$$180^\circ - 65^\circ = 115^\circ$$

Q4: Types of triangles

Acute, Right-angled, Obtuse

Q5: Sum of quadrilateral angles

$$360^\circ$$

Q6: Perimeter of rectangle

$$\text{Perimeter} = 2(l + w) = 2(8 + 5) = 26 \text{ cm}$$

Q7: Perpendicular lines

Two lines that meet at 90° angle

Q8: Equal angles by transversal

Corresponding angles or Alternate angles

SECTION B SOLUTIONS

Q9: Angles in ratio 2:3:3:4

Total parts = 12, Each angle: $2/12 \times 360^\circ = 60^\circ$, $3/12 \times 360^\circ = 90^\circ$, $3/12 \times 360^\circ = 90^\circ$, $4/12 \times 360^\circ = 120^\circ$

Q10: Triangle angles $2x$, $3x$, $4x$

$$2x + 3x + 4x = 180^\circ, 9x = 180^\circ, x = 20^\circ. \text{ Angles: } 40^\circ, 60^\circ, 80^\circ$$

Q11: Parallelogram angles

$70^\circ, 110^\circ, 70^\circ, 110^\circ$ (opposite angles equal, adjacent angles supplementary)

Q12: Square calculations

$$\text{Perimeter} = 4 \times 6 = 24 \text{ cm}, \text{ Area} = 6 \times 6 = 36 \text{ cm}^2$$

Q13: Angles on straight line

$(3x + 10) + (2x - 10) = 180^\circ$, $5x = 180^\circ$, $x = 36^\circ$. Angles: 118° and 62°

Q14: Isosceles triangle

$$180^\circ - 65^\circ - 65^\circ = 50^\circ$$

Q15: Vertically opposite angles

When two lines intersect, they form two pairs of vertically opposite angles. These are equal because they are formed by the same two intersecting lines.

GCSEMathsAI